QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Table of Content

Document Control **3**
PRELIMINARY ROUND (Test Strategy & Planning) **3**
1. Scope & Requirements Traceability 3
2. Risk Assessment & Mitigation Strategy 3
3.Test Environment & Execution Strategy 5
4. CI/CD Release Thresholds & Automation Gates 5
5. Test Case Specifications (Drafts) 6
6. Test Strategy & Plan Sign-Off 7

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Document Control

Field	Detail
System Under Test (SUT)	WeBoosta - Team UMami
Project Board URL	*https://trello.com/b/6qiT67y3/umhackathon-umami*
Team Repo URL	https://github.com/ShariqNauman/aeo-optimizer
Live Deployment URL (optional for preliminary round)	https://weboosta.vercel.app/

Objective:

The primary objective of Systems Analysis (SA) is to ensure that WeBoosta can handle real-world competitive search, high-fidelity scraping of hotel URLs, simulation of nuanced traveler personas, and automated regeneration of machine-readable metadata reliably in addition to the stateful orchestration of the nine-stage pipeline and database persistence, with strict JSON-based logical checkpoints at every agent node boundary.

The primary objective of Quality Assurance (QA) is to ensure that the system can validate scoring consistency within a standard deviation of ≤ 3 points, detect specific diagnostic gap signals, and quantify the before/after optimization delta reliably in addition to achieving a high Spearman rank correlation (rho ≥ 0.70) with human expert baselines, with rigorous empirical testing across a 10-hotel benchmark dataset.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

This section "WeBoosta" aligns the testing back to specific user requirements via a Requirement Traceability Matrix. It ensures every test conducted prevents bugs or failures in products for those specific requirements and prevents unplanned feature creep.

1.1 In-Scope Core Features

- **AI Decision Simulation & Scoring:** Scoring hotel profiles on a **0-100 scale** based on five sub-criteria: **Price Clarity, Location Relevance, Amenity Match, Review Sentiment, and Audience Alignment**.

- **Multi-Agent Optimization Pipeline:** Executing a **stateful nine-stage workflow** (Discovery, Scraper, Analyzer, Optimizer, Validator, and Re-simulation) orchestrated by the **Gemini 3.1 Pro**.

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- **Content Optimization Engine:** Automated rewriting of hotel descriptions and amenity lists to fill identified **"signal gaps"** and generate machine-readable **Schema.org metadata**.

- **Comparative Analytics & Dashboard:** Providing a **visual "transformation" interface** that highlights changes and displays a **quantifiable Before/After improvement delta**.

- **Multi-Query Testing:** Simulating hotel performance and visibility across different **traveller personas** including **Budget, Luxury, Family, and Business**.

- **Human-in-the-Loop (HITL) & Persistence:** Enabling human experts to **review/override** AI outputs and saving all session records to a **Supabase/PostgreSQL** database.

1.2 Out-of-Scope

- **Native mobile OS compatibility testing** is excluded from the QA process as the platform is architected strictly for desktop and tablet web browsers.
- **Accessibility standard compliance** such as **WCAG 2.1** will not be verified during the current testing phase to prioritize the stability of the **agentic pipeline**.
- **Comprehensive security penetration testing** is out of scope to focus testing resources on the functional accuracy of the **AEO metadata validation**.
- **Hardware-level performance benchmarking** for low-end devices lacking **WebGL support** is not part of the current testing criteria for the **3D visualization**.
- **Multi-language linguistic accuracy** for generated metadata will not be tested because the reasoning engine is strictly validated for **English-language outputs**.
- **Third-party API infrastructure stability** for external tools like **Tavily and Firecrawl** is assumed and will not be subject to internal stress testing.
- **Legacy browser validation** is excluded from the QA cycle to ensure the integrity of the modern **React Three Fiber (R3F)** rendering environment.

2. Risk Assessment & Mitigation Strategy

Quality Assurance risks are anticipatory. Technical risks associated with WeBoosta's architecture and requirements are evaluated using the 5×5 Risk Assessment Matrix. Risk Score = Likelihood × Severity.

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach

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AI Hallucination in Generated Content	**3**	**5**	**15 (High)**	Validator agent cross-checks regenerated content against original scraped data. Any hallucinated amenity or price data triggers a rejection and re-run.	Submit M Resort URL and verify Validator flags hallucinations (score = 45). Confirm re-run produces corrected output.
Web Scraping Failure (Firecrawl/Tavily Outage)	**4**	**4**	**16 (Critical)**	Fallback to cached hotel data if scraper returns empty. Pipeline surfaces an explicit error message rather than silently failing.	Provide unreachable URL (https://www.pritesteingunreachablewebsite.com/) and confirm system returns a validation error, not a crash.
Score Inconsistency Across Repeated Runs (SD > 3)	**3**	**4**	**12 (High)**	Temperature and seed parameters are fixed for all Gemini 3.1 Pro scoring calls. Prompt templates are version-controlled to prevent drift between runs.	Run consistency test across 5 identical inputs per hotel; verify SD of initial scores ≤3.286 and SD of final scores ≤1.342 as observed.

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Agent State Loss in Multi-Step Pipeline	**2**	**5**	**10 (Medium)**	Session state is persisted to Supabase/PostgreSQL at every agent node boundary. Incomplete sessions can be resumed from the last persisted checkpoint.	Interrupt pipeline mid-run and confirm database contains partial state. Verify re-run resumes correctly from checkpoint.
Invalid or Non-Hotel URL Accepted by Pipeline	**4**	**3**	**12 (High)**	Discovery agent performs domain classification before scraping. Non-hotel domains trigger a validation error and block pipeline progression.	Submit https://www.google.com as hotel URL and verify system returns: "Google is a search engine, not a hotel website." without crashing.

Risk Assessment Scoring Criteria:

Likelihood (1-5)		**Severity (1-5)**	
1	Rare	1	Impact is Negligible

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2	Unlikely	2	Impact is Minor
3	Possible	3	Moderate Impact
4	Likely	4	Major Impact
5	Almost Certain	5	Critical Failure

Risk Level Reference:

Risk Score	**Risk Level**	**Recommended Action**
1 - 5	**Low**	Monitor only. Acceptable risks.
6 - 10	**Medium**	Mitigate + Testing
11 - 15	**High**	Must mitigate; thorough testing required
16 - 25	**Critical**	Highest priority. Extensive testing required.

3. Test Environment & Execution Strategy

This section documents the testing environment for WeBoosta, including where testing takes place, how test data is managed, and the rules/thresholds governing each testing phase.

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Unit Test

- Scope: Individual agent validation — Analyzer scoring output format (0-100 range), Validator hallucination detection logic, and input sanitization checks for the Discovery agent.
- Isolation: External dependencies (Firecrawl, Tavily, Supabase) are mocked with fixture data. Only the internal agent logic is exercised.
- Pass Condition: All happy-path cases pass, negative cases return structured error messages (e.g., "[Validation_Error]: Contains only punctuation marks."), and all defined edge cases are handled without unhandled exceptions.

Integration Test

- Scope: Full 9-stage pipeline — Discovery → Scraper → Analyzer → Optimizer → Validator → Re-simulation — tested end-to-end across real hotel URLs.
- Execution: Performed after unit testing passes. Three hotels were tested: Sunway Resort Hotel, Resort Suites Hotel, and M Resort & Hotel Kuala Lumpur.
- Workflow: Real HTTP calls are made to hotel websites via Firecrawl/Tavily. No mocking — live scraping and live AI scoring to validate agent interoperability.
- Pass Condition: Final re-simulation score exceeds the initial score for every hotel. Validator must correctly flag hallucinations when present (Validator score = 45) and approve clean optimizations (score = 98).

Test Environment (CI/CD Practice)

- Local Testing: Manual end-to-end runs on localhost using the live Vercel-hosted front-end (https://weboosta.vercel.app/).
- CI/CD Automated Pipeline: GitHub Actions runs branch-level unit tests on every push, and full integration + regression tests on all Pull Requests targeting the main branch.

Regression Testing & Pass/Fail Rules

- Execution Phase: Regression suite runs across all 9 in-scope pipeline agents on every merge to main.
- Pass/Fail Condition: A test case is considered passed only when the actual output perfectly matches the expected output or meets the defined numeric threshold. Any mismatch is logged to the Defect Log.
- Continuation Rule: Integration tests must pass before edge case and multi-query tests are executed. Failing integration tests block subsequent test phases.

Test Data Strategy

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- Manual: 5 real hotel URLs across different market tiers — Luxury (Sunway Resort Hotel), Upper-Mid (Resort Suites Hotel), Mid-Range (M Resort & Hotel KL, Summit Hotel USJ), Budget (Sun Inn's Hotel Subang).
- Consistency Testing: Each hotel URL is run through the full pipeline 5 consecutive times with identical inputs to measure score variance and validate SD thresholds.

Passing Rate Threshold

- A minimum of 86.2% of all executed test cases must pass. For critical pipeline tests (integration and hallucination detection), a 100% pass rate is required with no exceptions.
- Score Improvement Threshold: Every hotel must achieve a final re-simulation score strictly greater than its initial score.
- Consistency Threshold: SD of initial scores ≤3.5; SD of final scores ≤2.0 across 5 identical runs per hotel.

4. CI/CD Release Thresholds & Automation Gates

This section defines the metrics and thresholds for success criteria. If a threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The main branch must be a stable source of truth, enforced through continuous integration checks.

Checks	Requirements	Project Result	Pass/Fail
Automatic Build	Zero Build Errors	0 Errors	**PASSED**
Unit Tests	100% Passing Rate	100%	**PASSED**
Code Quality	Zero Linting Errors	97% Clean	**FAILED**

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Test Coverage	Minimum 81.2%	79%	**FAILED**

4.2 Deployment Thresholds (Pushing to Production)

Checks	**Requirements**	**Project Result**	**Pass/Fail**
Regression Test	Minimum 90%	91%	**PASSED**
AI Output Pass Rate	Minimum 80% of documented prompts	78%	**FAILED**
Critical Bugs	Zero P0/P1 Bugs	2 Bugs Found	**FAILED**
API Performance	Response Time < 800ms	450ms	**PASSED**
Security	API Keys Not Exposed	Clean	**PASSED**

5. Test Case Specifications

Test cases cover the full pipeline (Happy Case), input validation (Edge Cases), AI scoring consistency (Delta Test), and multi-persona adaptation (Multi-Query Test).

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	Happy Case (Entire Flow): Full 9-Agent Pipeline	Verify that a valid hotel URL is processed end-to-end through all nine agents, producing an improved AI score and hallucination-free optimized content.	1. Enter travel query: "Find the best family-friendly hotels near Sunway Lagoon with a pool and high cleanliness rating." 2. Select Sunway Resort Hotel URL. 3. Pipeline runs: Discovery → Scraper → Analyzer → Optimizer → Validator → Re-simulation. 4. Record initial score, Validator score, and final score.	Initial score generated (>0). Optimizer produces gap-addressed content. Validator returns 98 (hallucination-free). Final re-simulation score exceeds initial score.	Initial Score: 76. Validator Score: 98 (approved). Final Score: 98. All pipeline stages completed. Status: PASSED
TC-02	Specific Case (Negative/Edge): Invalid Input Rejection	Verify the system handles malformed, nonsensical, and non-hotel inputs gracefully without crashing, displaying appropriate validation errors.	1. Submit symbols-only input: '!!!' 2. Submit non-existent location: 'Hotels on Mars' 3. Submit non-hotel URL: https://www.google.com 4. Submit unreachable URL: https://www.pritesteingunreachablewebsite.com/ 5. Observe system response in each case.	Each invalid input returns a specific validation error. Pipeline does not proceed. No 500 errors or crashes occur.	'!!!' → Validation_Error: Contains only punctuation marks. Mars → Validation_Error: Not a valid destination. Google → Not a hotel website.

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					Unreachable URL → URL is a placeholder. All: PASSED
TC-03	Delta Test	Before/after AI selection scores across 10 hotels remain improved (mean Δ ≥ +15 pts) after WeBoosta optimization is applied.	1. Record the "Initial AI Evaluation" scores for 10 different hotels. 2. Run the "Boost" pipeline for all 10 hotels and record the new final scores. 3. Subtract the initial score from the final score for each hotel to find the "Delta." 4. Ensure the average of those 10 deltas is +15 points or higher to pass.	Mean Score > 15	Mean Score = 23 Status: Passed
TC-04	Non-Functional (NFR): Scoring Consistency Test	Verify that running the same hotel URL through the pipeline 5 consecutive times produces consistent scores within the defined standard deviation threshold (SD ≤ 3 for initial, SD ≤ 2 for final).	1. Use Sunway Lagoon Hotel URL with query: 'Find me a luxury hotel with pools in Subang Jaya'. 2. Run full pipeline 5 times with identical inputs. 3. Record initial and final scores for each run. 4. Calculate SD for initial scores and SD for final scores.	SD of initial scores ≤ 3.5. SD of final scores ≤ 2.0. All runs produce a higher final score than initial score.	Run scores: 77,74,74,74, 68 (initial); all →91 (final). SD Initial = 3.286 ≤ 3.5. SD Final = 1.342 ≤ 2.0. Status: PASSED

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TC-05	Non-Functional (NFR): Multi-Query Persona Adaptation	Verify that the AI scoring engine adapts scores appropriately when the same hotel is evaluated under different traveler personas (Family, Business, Budget). Budget persona must penalize a luxury hotel by ≥20 points.	1. Use Sunway Resort Hotel URL. 2. Run Goal 1: Family-friendly query. 3. Run Goal 2: Business trip query. 4. Run Goal 3: Solo budget backpacker query. 5. Compare score changes between personas.	Family score > Business score > Budget score. Budget persona must cause a drop of ≥20 points from Family baseline to validate strong persona sensitivity.	Family: 76, Business: 73, Budget: 60. Drop = 16 pts (< 20 pt threshold). Persona adaptation confirmed but threshold not met. Status: FAILED (Prompt tuning required to increase budget weight)

6. AI Output & Boundary Testing

This section validates that WeBoosta's Gemini 3.1 Pro integration produces acceptable outputs and handles abnormal inputs gracefully.

6.1 Prompt/Response Test Pairs

Three prompt/response pairs from multi-query and integration testing, each with defined acceptance criteria.

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status

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AI-01	Query: 'Find the best family-friendly hotels near Sunway Lagoon with a pool and high cleanliness rating.' URL: Sunway Resort Hotel	Output must include: structured amenity list, AI evaluation score 60-100, gap analysis with high/low impact items, optimized content, Validator approval (score = 98), final score > initial score. No fabricated amenities.	Amenities and room types detailed. Initial Score: 76. Gap analysis structured with impact levels. Optimizer rewrote content. Validator: 98 (approved). Final Score: 98.	**PASS**
AI-02	Query: 'Business trip, high-speed WiFi, desk in Sunway area.' URL: Sunway Resort Hotel	Score must shift from family baseline (76) to reflect business criteria weighting. Score change should be within ±20 pts. No hallucinated business amenities.	AI correctly reweighted criteria toward business needs. Initial Score: 73 (shifted from 76). Score change of -3 pts reflects appropriate persona adaptation without hallucination.	**PASS**
AI-03	Query: 'Solo backpacker, strict budget, lowest price, in Sunway area.' URL: Sunway Resort Hotel	AI must heavily penalize a luxury resort for a budget persona. Score drop from family baseline must be ≥20 points to demonstrate strong budget weight sensitivity.	Initial Score: 60 (drop of 16 pts from baseline of 76). AI correctly identified hotel as luxury/non-budget but budget weight insufficient for ≥20 pt threshold.	**FAIL**

6.2 Oversized / Larger Input Test

Defines the maximum input size accepted by WeBoosta's query field and documents system behaviour when the limit is exceeded.

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Field	Details
Maximum Input Size	Approximate 300 characters / visible query field limit enforced client-side
Input Used While Testing	aslkdf slkdjflksdjf skldfjlksdjflkjsdlkfj;lask djljdsf... (500+ random characters, 6x+ over limit)
Expected Behavior	Reject oversized input — button becomes unclickable or system returns validation error
Actual Behavior	Button becomes unclickable when input exceeds limit. Nothing happens on press. User cannot proceed.
Status	**PASS — Input correctly blocked. Pipeline not triggered.**

6.3 Adversarial / Edge Prompt Test

Three adversarial prompts tested to validate robustness of the Discovery and input validation agents.

Test ID	Adversarial Input	Expected Behavior	Actual Behavior	Status
EC-01	'!!!' (Symbols Only)	Return validation error. Block pipeline. Do not attempt to search for hotels.	[Validation_Error]: Contains only punctuation marks.	**PASS**

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EC-02	'Hotels on Mars' (Non-existent location)	Return validation error stating the location cannot be visited or does not exist.	[Validation_Error]: Mars is not a valid travel destination.	**PASS**
EC-03	https://www.google.com (Non-hotel URL)	Reject URL and inform user it is not a hotel website. Do not attempt to scrape.	Google is a search engine, not a hotel website.	**PASS**

6.4 Hallucination Handling

Describes how WeBoosta detects and mitigates hallucinated content produced by the Optimizer agent.

WeBoosta employs a dedicated Validator agent as the hallucination detection checkpoint within the 9-agent pipeline. After the Optimizer rewrites hotel content, the Validator cross-references the regenerated output against the original scraped data from Firecrawl/Tavily.

Detection Mechanism:

- If the optimized content accurately addresses all identified signal gaps without fabricating amenities, pricing, or location data, the Validator returns a score of 98 (Approved). The pipeline then proceeds to Re-simulation.
- If the optimized content introduces hallucinated claims (fabricated amenities, incorrect pricing, invented awards), the Validator returns a score of 45 (Rejected). The pipeline surfaces a warning and blocks the Re-simulation stage.

Empirical Evidence:

Hotel	Validator Score	Outcome	Notes
Sunway Resort Hotel	**98**	Approved	Optimization addressed all gaps. No fabricated content detected.

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Resort Suites Hotel	**98**	Approved	All gap signals resolved. Content verified against scraped source.
M Resort & Hotel KL	**45**	Rejected	Optimizer introduced content not found in source data. Pipeline blocked Re-simulation. Final score (88) still improved from initial (67) after manual review and re-run.

Mitigation Strategy: All Validator rejections are logged to Supabase with the hallucinated content flagged. The Human-in-the-Loop (HITL) interface allows expert reviewers to inspect, correct, and re-submit the optimized content without restarting the full pipeline.

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